

Import Libraries

```
import pandas as pd
import numpy as np
```

Load Dataset

```
data = pd.read_csv("/content/Titanic_Dataset.csv")
```

```
data.shape
```

```
(891, 12)
```

```
data.columns
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

```
print(data.head())
```

```

 PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

      Name               Sex  Age  SibSp  \
0  Braund, Mr. Owen Harris   male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0    1
2    Heikkinen, Miss. Laina  female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0    1
4    Allen, Mr. William Henry   male  35.0    0

   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN      S
1      0    PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282  7.9250   NaN      S
3      0  113803  53.1000  C123      S
4      0  373450   8.0500   NaN      S
```

Checking the dataset information

```
print(data.info())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype  
---  --
0   PassengerId    891 non-null    int64  
1   Survived       891 non-null    int64  
2   Pclass         891 non-null    int64  
3   Name           891 non-null    object  
4   Sex            891 non-null    object  
5   Age            714 non-null    float64 
6   SibSp          891 non-null    int64  
7   Parch          891 non-null    int64  
8   Ticket         891 non-null    object  
9   Fare           891 non-null    float64 
10  Cabin          204 non-null    object  
11  Embarked       889 non-null    object  
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

```
print(data.isnull().sum())
```

```

PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
```

```
Cabin      687
Embarked    2
dtype: int64
```

Handling Missing Values

```
data['Age'] = data['Age'].fillna(data['Age'].mean())
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])

data.drop(columns=['Cabin'], inplace = True)
```

```
print(data.isnull().sum())
```

```
PassengerId    0
Survived        0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

Convert Text Data into Numbers

```
from sklearn.preprocessing import LabelEncoder
```

```
le = LabelEncoder()
data['Sex'] = le.fit_transform(data['Sex'])
data['Embarked'] = le.fit_transform(data['Embarked'])
```

Select Features and Target

```
X = data[['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked']]
y = data['Survived']
```

Split Dataset

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 42)
```

Creating Machine Learning Model

```
from sklearn.linear_model import LogisticRegression
```

```
model = LogisticRegression()
```

Training Model

```
model.fit(X_train, y_train)
```

```
▼ LogisticRegression ⓘ ?
LogisticRegression()
```

Predicting Result

```
y_pred = model.predict(X_test)
```

Checking Accuracy

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:",accuracy)
```

Accuracy: 0.8100558659217877

Confusion Matrix

```
confusion_matrix = confusion_matrix(y_test,y_pred)
print("Confusion Matrix:")
print(confusion_matrix)
```

Confusion Matrix:
[[90 15]
 [19 55]]

Survival Count where 0 represents Not Survived and 1 represents Survived

```
survival_count = data['Survived'].value_counts()

print("\nSurvival Count:")
print(survival_count)

# Explain values
print("\nExplanation:")
print("0 = Not Survived")
print("1 = Survived")
```

Survival Count:
Survived
0 549
1 342
Name: count, dtype: int64

Explanation:
0 = Not Survived
1 = Survived

Graph Representation

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(3,3))
sns.countplot(x = 'Survived',data = data)
plt.title("Titanic Survival Count")
plt.xlabel("Survived")
plt.ylabel("Count")
plt.show()
```



